# Ye Yuan

Staff Research Scientist & Tech Lead, NVIDIA Research

**ROBOT LEARNING / HUMANOIDS / EMBODIED AI**

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## Research profile

Research scientist and technical leader with extensive experience in reinforcement learning and robotics. Develops humanoid learning systems that connect reinforcement learning, visuomotor control, human data, and generative models. Research spans foundational algorithms, data collection and generation, large-scale training, and real-robot evaluation. Record includes 47 accepted or published conference/journal papers, nine oral/highlight/spotlight selections, and a SIGGRAPH Best Paper Honorable Mention.

## Selected robotics research contributions

**Scalable reinforcement learning and whole-body control**

Co-first author of SONIC (*Science Robotics*, 2026): scaled motion tracking to 42M parameters, 100M+ motion frames, and 21K GPU-hours, with a shared control interface for teleoperation and vision-language-action models. Widely adopted by robotics companies and researchers. Experience with whole-body teleoperation, UMI (Universal Manipulation Interface) data collection, and training System 0 and System 1 UMI policies.

**Visuomotor learning and sim-to-real transfer**

Contributed to VIRAL (CVPR 2026), scaling visual teacher-student training to 64 GPUs and demonstrating 54 continuous zero-shot loco-manipulation cycles on a physical Unitree G1. Related visual RL research studies active perception and dexterity in simulated humanoids. First-author work (ECCV 2018; ICCV 2019) connected egocentric vision to physics-based humanoid control and motion forecasting through imitation learning and RL.

**Learning robot skills from human video and world models**

Project lead for GRAIL (CoRL 2026 Oral), using video foundation models in fully specified 3D scenes to generate 20K+ humanoid demonstrations. Egocentric RGB policies trained solely on GRAIL-generated data achieved 84% pick-up success on seen objects, 80% on unseen objects, and 90% stair-climbing success on a real Unitree G1.

**Generative motion and 3D foundations**

Main model architect for Kimodo, a controllable motion diffusion model trained on 700 hours of motion capture. First author of PhysDiff (ICCV 2023 Oral) and last author of GEM (formerly GENMO) (ICCV 2025 Highlight). Co-first author of GAvatar; additional 3D generation work includes AGG and AdaHuman.

## Research experience

**NVIDIA Research** May 2022 – Present

*Staff Research Scientist & Tech Lead, DAIR and LPR*

Lead research connecting human data, simulation, and generative models to humanoid control and visual policies. Guide collaborations from problem formulation and model design through training, open-source release, and real-robot evaluation.

**Carnegie Mellon University, Robotics Institute** Aug 2017 – May 2022

*Ph.D. Research Assistant; advisor: Kris Kitani*

Developed reinforcement-learning methods for humanoid control (Residual Force Control, NeurIPS 2020), visuomotor control, and automated robot design algorithms (Transform2Act, ICLR 2022 Oral). Mentored graduate and undergraduate researchers.

## Honors

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| SIGGRAPH Best Paper Honorable Mention | 2023 |
| NVIDIA Graduate Fellowship (1 of 5 selected worldwide) | 2021–2022 |
| Qualcomm Innovation Fellowship ($100K award) | 2020–2021 |
| Outstanding Reviewer, ECCV / ICLR | 2022 / 2021 |

## Selected publications

*Co-first/equal contribution; †project lead. Complete record: appendix.

### Humanoid control and reinforcement learning

1. SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control
   Z. Luo*, **Y. Yuan***, T. Wang*, et al. | Science Robotics 2026
2. Transform2Act: Learning a Transform-and-Control Policy for Efficient Agent Design
   **Y. Yuan**, Y. Song, Z. Luo, W. Sun, K. Kitani | ICLR 2022; Oral
3. Residual Force Control for Agile Human Behavior Imitation and Extended Motion Synthesis
   **Y. Yuan**, K. Kitani | NeurIPS 2020

### Visuomotor learning and sim-to-real

4. VIRAL: Visual Sim-to-Real at Scale for Humanoid Loco-Manipulation
   T. He*, Z. Wang*, H. Xue*, …, **Y. Yuan**, et al. | CVPR 2026
5. Emergent Active Perception and Dexterity of Simulated Humanoids from Visual Reinforcement Learning
   Z. Luo, C. Tessler, T. Lin, **Y. Yuan**, et al. | arXiv 2025
6. Ego-Pose Estimation and Forecasting as Real-Time PD Control
   **Y. Yuan**, K. Kitani | ICCV 2019
7. 3D Ego-Pose Estimation via Imitation Learning
   **Y. Yuan**, K. Kitani | ECCV 2018

### Human data for robot learning

8. GRAIL: Generating Humanoid Loco-Manipulation from World Models
   T. Xie*, H. Zhang*, J. Park*, …, **Y. Yuan**† | CoRL (accepted) 2026; Oral
9. GLAMR: Global Occlusion-Aware Human Mesh Recovery with Dynamic Cameras
   **Y. Yuan**, U. Iqbal, P. Molchanov, K. Kitani, J. Kautz | CVPR 2022; Oral
10. SimPoE: Simulated Character Control for 3D Human Pose Estimation
    **Y. Yuan**, S.-E. Wei, T. Simon, K. Kitani, J. Saragih | CVPR 2021; Oral

### Generative motion and 3D modeling

11. Kimodo: Scaling Controllable Human Motion Generation
    D. Rempe*, M. Petrovich*, **Y. Yuan**, et al. | arXiv 2026
12. AdaHuman: Animatable Detailed 3D Human Generation with Compositional Multiview Diffusion
    Y. Huang, **Y. Yuan**, X. Li, J. Kautz, U. Iqbal | ICCV 2025
13. GAvatar: Animatable 3D Gaussian Avatars with Implicit Mesh Learning
    **Y. Yuan***, X. Li*, Y. Huang, S. De Mello, K. Nagano, J. Kautz, U. Iqbal | CVPR 2024; Highlight
14. PhysDiff: Physics-Guided Human Motion Diffusion Model
    **Y. Yuan**, J. Song, U. Iqbal, A. Vahdat, J. Kautz | ICCV 2023; Oral

## Education

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|---|---|
| **Carnegie Mellon University**, Ph.D. Robotics; advisor: Kris Kitani | 2022 |
| **Carnegie Mellon University**, M.S. Computer Science; advisor: Stelian Coros | 2016 |
| **Zhejiang University**, B.E. Computer Science and Technology | 2015 |

## Research service

**Area Chair:** CVPR; **Reviewer:** NeurIPS, ICML, ICLR, CVPR, ICCV, ECCV, ICRA, SIGGRAPH, JMLR, TMLR, TPAMI, TIP, RA-L.

## Research software and systems

**Project releases (GitHub stars):** SONIC / GR00T-WBC (3,773), GRAIL (575), GEM/GENMO (528), GEM-X (484).
**Personal projects:** PyTorch-RL (1,286), AgentFormer (317), RFC (169), DLow (125). As of October 17, 2026.
**Systems:** Python, PyTorch, distributed GPU training, Isaac Gym / Isaac Sim, MuJoCo, Unitree G1.

## Complete publication record

47 accepted or published conference/journal papers; 9 oral/highlight/spotlight selections. Preprints, the workshop paper, and the thesis are listed separately. Conference and journal versions of the same work are counted once. Contribution symbols follow each paper's own legend.

## Conference and journal papers

1. **SONIC: Supersizing Motion Tracking for Natural Humanoid Whole-Body Control**
   Zhengyi Luo[†], **Ye Yuan**[†], Tingwu Wang[†], Chenran Li[†], Fernando Castañeda[†], Sirui Chen[*], Zi-Ang Cao[*], Jiefeng Li[*], David Minor[*], Qingwei Ben[*], Jinhyung Park[*], David Sami[*], Zi Wang[*], Xingye Da[*], Runyu Ding, Cyrus Hogg, Lina Song, Edy Lim, Eugene Jeong, Tairan He, Haoru Xue, Wenli Xiao, Simon Yuen, Jan Kautz, Yan Chang, Umar Iqbal, Linxi "Jim" Fan[‡], Yuke Zhu[‡].
   *Science Robotics 2026*. [†]Co-first authors; *Core contributors; [‡]Equal advising.

2. **GRAIL: Generating Humanoid Loco-Manipulation from World Models**
   Tianyi Xie[*], Haotian Zhang[*], Jinhyung Park[*], Zi Wang[*], Bowen Wen, Jiefeng Li, Xueting Li, Qingwei Ben, Haoyang Weng, Yufei Ye, David Minor, Tingwu Wang, Chenfanfu Jiang, Sanja Fidler, Jan Kautz, Linxi "Jim" Fan, Yuke Zhu, Zhengyi Luo[†], Umar Iqbal[†], **Ye Yuan**[†].
   *CoRL (accepted) 2026; Oral*. *Co-First Authors, [†]Project Leads.

3. **MotionBricks: Scalable Real-Time Motions with Modular Latent Generative Model and Smart Primitives**
   Tingwu Wang[*], Olivier Dionne[*], Michael De Ruyter, David Minor, Davis Rempe, Kaifeng Zhao, Mathis Petrovich, **Ye Yuan**, Chenran Li, Zhengyi Luo, Brian Robison, Xavier Blackwell, Bernardo Antoniazzi, Xue Bin Peng, Yuke Zhu, Simon Yuen.
   *ACM TOG / SIGGRAPH 2026*. *Joint first authors.

4. **VIRAL: Visual Sim-to-Real at Scale for Humanoid Loco-Manipulation**
   Tairan He[*], Zi Wang[*], Haoru Xue[*], Qingwei Ben[*], Zhengyi Luo, Wenli Xiao, **Ye Yuan**, Xingye Da, Fernando Castañeda, Shankar Sastry, Changliu Liu, Guanya Shi, Linxi "Jim" Fan[†], Yuke Zhu[†].
   *CVPR 2026*. *Equal Contribution, [†]Project Leads.

5. **CARI4D: Category Agnostic 4D Reconstruction of Human-Object Interaction**
   Xianghui Xie, Bowen Wen, Yan Chang, Hesam Rabeti, Jiefeng Li, **Ye Yuan**, Gerard Pons-Moll, Stan Birchfield.
   *CVPR 2026*.

6. **Dream, Lift, Animate: From Single Images to Animatable Gaussian Avatars**
   Marcel C. Bühler, **Ye Yuan**, Xueting Li, Yangyi Huang, Koki Nagano, Umar Iqbal.
   *3DV 2026*.

7. **GENMO: A GENeralist Model for Human MOtion**
   Jiefeng Li, Jinkun Cao, Haotian Zhang, Davis Rempe, Jan Kautz, Umar Iqbal, **Ye Yuan**.
   *ICCV 2025; Highlight*.

8. **AdaHuman: Animatable Detailed 3D Human Generation with Compositional Multiview Diffusion**
   Yangyi Huang, **Ye Yuan**, Xueting Li, Jan Kautz, Umar Iqbal.
   *ICCV 2025*.

9. **GeoMan: Temporally Consistent Human Geometry Estimation using Image-to-Video Diffusion**
   Gwanghyun Kim, Xueting Li, **Ye Yuan**, Koki Nagano, Tianye Li, Jan Kautz, Se Young Chun, Umar Iqbal.
   *ICCV 2025*.

10. **SimAvatar: Simulation-Ready Avatars with Layered Hair and Clothing**
    Xueting Li, **Ye Yuan**, Shalini De Mello, Gilles Daviet, Jonathan Leaf, Miles Macklin, Jan Kautz, Umar Iqbal.
    *CVPR 2025*.

11. **BLADE: Single-view Body Mesh Estimation through Accurate Depth Estimation**
    Shengze Wang, Jiefeng Li, Tianye Li, **Ye Yuan**, Henry Fuchs, Koki Nagano, Shalini De Mello, Michael Stengel.
    *CVPR 2025*.

12. **Harmon: Whole-Body Motion Generation of Humanoid Robots from Language Descriptions**
    Zhenyu Jiang[*], Yuqi Xie[*], Jinhan Li, **Ye Yuan**, Yifeng Zhu, Yuke Zhu.
    *CoRL 2024*. *Equal contribution.

13. **COIN: Control-Inpainting Diffusion Prior for Human and Camera Motion Estimation**
Jiefeng Li, **Ye Yuan**, Davis Rempe, Haotian Zhang, Pavlo Molchanov, Cewu Lu, Jan Kautz, Umar Iqbal.
*ECCV 2024*.

14. **AGG: Amortized Generative 3D Gaussians for Single Image to 3D**
Dejia Xu, **Ye Yuan**, Morteza Mardani, Sifei Liu, Jiaming Song, Zhangyang Wang, Arash Vahdat.
*TMLR 2024*.

15. **GAvatar: Animatable 3D Gaussian Avatars with Implicit Mesh Learning**
**Ye Yuan**[*], Xueting Li[*], Yangyi Huang, Shalini De Mello, Koki Nagano, Jan Kautz, Umar Iqbal.
*CVPR 2024; Highlight*. *Equal contribution.

16. **PACER+: On-Demand Pedestrian Animation Controller in Driving Scenarios**
Jingbo Wang[*], Zhengyi Luo[*], **Ye Yuan**, Yixuan Li, Bo Dai.
*CVPR 2024*. *Equal contribution.

17. **PACE: Human and Camera Motion Estimation from in-the-wild Videos**
Muhammed Kocabas, **Ye Yuan**, Pavlo Molchanov, Yunrong Guo, Michael J. Black, Otmar Hilliges, Jan Kautz, Umar Iqbal.
*3DV 2024; Spotlight*.

18. **PhysDiff: Physics-Guided Human Motion Diffusion Model**
**Ye Yuan**, Jiaming Song, Umar Iqbal, Arash Vahdat, Jan Kautz.
*ICCV 2023; Oral*.

19. **Learning Human Dynamics in Autonomous Driving Scenarios**
Jingbo Wang, **Ye Yuan**, Zhengyi Luo, Kevin Xie, Dahua Lin, Umar Iqbal, Sanja Fidler, Sameh Khamis.
*ICCV 2023*.

20. **Learning Physically Simulated Tennis Skills from Broadcast Videos**
Haotian Zhang, **Ye Yuan**, Viktor Makoviychuk, Yunrong Guo, Sanja Fidler, Xue Bin Peng, Kayvon Fatahalian.
*ACM TOG / SIGGRAPH 2023; Best Paper Honorable Mention*.

21. **Trace and Pace: Controllable Pedestrian Animation via Guided Trajectory Diffusion**
Davis Rempe[*], Zhengyi Luo[*], Xue Bin Peng, **Ye Yuan**, Kris Kitani, Karsten Kreis, Sanja Fidler, Or Litany.
*CVPR 2023*. *Equal contribution.

22. **RGB-Only Reconstruction of Tabletop Scenes for Collision-Free Manipulator Control**
Zhenggang Tang, Balakumar Sundaralingam, Jonathan Tremblay, Bowen Wen, **Ye Yuan**, Stephen Tyree, Charles Loop, Alexander Schwing, Stan Birchfield.
*ICRA 2023*.

23. **Embodied Scene-aware Human Pose Estimation**
Zhengyi Luo[*], Shun Iwase[*], **Ye Yuan**, Kris Kitani.
*NeurIPS 2022*. *Equal contribution.

24. **GLAMR: Global Occlusion-Aware Human Mesh Recovery with Dynamic Cameras**
**Ye Yuan**, Umar Iqbal, Pavlo Molchanov, Kris Kitani, Jan Kautz.
*CVPR 2022; Oral*.

25. **Transform2Act: Learning a Transform-and-Control Policy for Efficient Agent Design**
**Ye Yuan**, Yuda Song, Zhengyi Luo, Wen Sun, Kris Kitani.
*ICLR 2022; Oral*.

26. **Online No-regret Model-Based Meta RL for Personalized Navigation**
Yuda Song, **Ye Yuan**, Wen Sun, Kris Kitani.
*L4DC 2022*.

27. **Patcher: Patch Transformers with Mixture of Experts for Precise Medical Image Segmentation**
Yanglan Ou, **Ye Yuan**, Xiaolei Huang, Stephen T. C. Wong, John Volpi, James Z. Wang, Kelvin Wong.
*MICCAI 2022*.

28. **Dynamics-Regulated Kinematic Policy for Egocentric Pose Estimation**
Zhengyi Luo, Ryo Hachiuma, **Ye Yuan**, Kris Kitani.
*NeurIPS 2021*.

29. **AgentFormer: Agent-Aware Transformers for Socio-Temporal Multi-Agent Forecasting**
**Ye Yuan**, Xinshuo Weng, Yanglan Ou, Kris Kitani.
*ICCV 2021*.

30. **SimPoE: Simulated Character Control for 3D Human Pose Estimation**
**Ye Yuan**, Shih-En Wei, Tomas Simon, Kris Kitani, Jason Saragih.
*CVPR 2021; Oral*.

31. **PTP: Parallelized Tracking and Prediction with Graph Neural Networks and Diversity Sampling**
Xinshuo Weng*, **Ye Yuan***, Kris Kitani.
*RA-L and ICRA 2021; Best Student Paper Award Candidate, ICRA.* *Equal Contribution.

32. **LambdaUNet: 2.5D Stroke Lesion Segmentation of Diffusion-weighted MR Images**
Yanglan Ou, **Ye Yuan**, Xiaolei Huang, Kelvin Wong, John Volpi, James Z. Wang, Stephen T. C. Wong.
*MICCAI 2021*.

33. **Residual Force Control for Agile Human Behavior Imitation and Extended Motion Synthesis**
**Ye Yuan**, Kris Kitani.
*NeurIPS 2020*.

34. **DLow: Diversifying Latent Flows for Diverse Human Motion Prediction**
**Ye Yuan**, Kris Kitani.
*ECCV 2020*.

35. **Efficient Non-Line-of-Sight Imaging from Transient Sinograms**
Mariko Isogawa, Dorian Yao Chan, **Ye Yuan**, Kris Kitani, Matthew O'Toole.
*ECCV 2020*.

36. **Diverse Trajectory Forecasting with Determinantal Point Processes**
**Ye Yuan**, Kris Kitani.
*ICLR 2020*.

37. **Optical Non-Line-of-Sight Physics-based 3D Human Pose Estimation**
Mariko Isogawa, **Ye Yuan**, Matthew O'Toole, Kris Kitani.
*CVPR 2020*.

38. **Generative Hybrid Representations for Activity Forecasting with No-Regret Learning**
Jiaqi Guan, **Ye Yuan**, Kris Kitani, Nick Rhinehart.
*CVPR 2020; Oral*.

39. **Back-Hand-Pose: 3D Hand Pose Estimation for a Wrist-worn Camera via Dorsum Deformation Network**
Erwin Wu, **Ye Yuan**, Hui-Shyong Yeo, Aaron Quigley, Hideki Koike, Kris Kitani.
*UIST 2020*.

40. **MonoEye: Multimodal Human Motion Capture System Using A Single Ultra-Wide Fisheye Camera**
Dong-Hyun Hwang, Kohei Aso, **Ye Yuan**, Kris Kitani, Hideki Koike.
*UIST 2020*.

41. **Semi-Supervised Cervical Dysplasia Classification With Learnable Graph Convolutional Network**
Yanglan Ou, Yuan Xue, **Ye Yuan**, Tao Xu, Vincent Pisztora, Jia Li, Xiaolei Huang.
*ISBI 2020*.

42. **Ego-Pose Estimation and Forecasting as Real-Time PD Control**
**Ye Yuan**, Kris Kitani.
*ICCV 2019*.

43. **3D Ego-Pose Estimation via Imitation Learning**
**Ye Yuan**, Kris Kitani.
*ECCV 2018*.

44. **Computational Design of Transformables**
**Ye Yuan**, Changxi Zheng, Stelian Coros.
*SCA 2018*.

45. **Interactive Co-Design of Form and Function for Legged Robots using the Adjoint Method**
Ruta Desai, Beichen Li, **Ye Yuan**, Stelian Coros.
*CLAWAR 2018*.

46. **Computational Abstractions for Interactive Design of Robotic Devices**
Ruta Desai, **Ye Yuan**, Stelian Coros.
*ICRA 2017*.

47. **Continuous Optimization of Interior Carving in 3D Fabrication**
Yue Xie, **Ye Yuan**, Xiang Chen, Changxi Zheng, Kun Zhou.
*Frontiers of Computer Science 2017*.

## Preprints and technical reports

1. **Kimodo: Scaling Controllable Human Motion Generation**
Davis Rempe[*], Mathis Petrovich[*], **Ye Yuan**, Haotian Zhang, Xue Bin Peng, Yifeng Jiang, Tingwu Wang, Umar Iqbal, David Minor, Michael de Ruyter, Jiefeng Li, Chen Tessler, Edy Lim, Eugene Jeong, Sam Wu, Ehsan Hassani, Michael Huang, Jin-Bey Yu, Chaeyeon Chung, Lina Song, Olivier Dionne, Jan Kautz, Simon Yuen, Sanja Fidler.
*arXiv 2026*. *Equal Contribution.

2. **CHIP: Adaptive Compliance for Humanoid Control through Hindsight Perturbation**
Sirui Chen[*], Zi-Ang Cao[*], Zhengyi Luo, Fernando Castañeda, Chenran Li, Tingwu Wang, **Ye Yuan**, Linxi "Jim" Fan, C. Karen Liu[†], Yuke Zhu[†].
*arXiv 2025*. *Equal Contribution, [†]Project Leads.

3. **Emergent Active Perception and Dexterity of Simulated Humanoids from Visual Reinforcement Learning**
Zhengyi Luo, Chen Tessler, Toru Lin, **Ye Yuan**, Tairan He, Wenli Xiao, Yunrong Guo, Gal Chechik, Kris Kitani, Jim Fan, Yuke Zhu.
*arXiv 2025*.

4. **SMPLOlympics: Sports Environments for Physically Simulated Humanoids**
Zhengyi Luo, Jiashun Wang, Kangni Liu, Haotian Zhang, Chen Tessler, Jingbo Wang, **Ye Yuan**, Jinkun Cao, Zihui Lin, Fengyi Wang, Jessica Hodgins, Kris Kitani.
*arXiv 2024*.

5. **From Universal Humanoid Control to Automatic Physically Valid Character Creation**
Zhengyi Luo, **Ye Yuan**, Kris M. Kitani.
*arXiv 2022*.

6. **All-In-One Drive: A Comprehensive Perception Dataset with High-Density Long-Range Point Clouds**
Xinshuo Weng, Yunze Man, Jinhyung Park, **Ye Yuan**, Matthew O'Toole, Kris Kitani.
*Technical report 2022*.

## Workshop paper

1. **Interacting with Intelligent Characters in AR**
Gokçen Çimen, **Ye Yuan**, Robert W. Sumner, Stelian Coros, Martin Guay.
*AIVRAR workshop 2017*.

## Doctoral thesis

1. **Unified Simulation, Perception, and Generation of Human Behavior**
**Ye Yuan**.
*Ph.D. thesis, Carnegie Mellon University 2022*.